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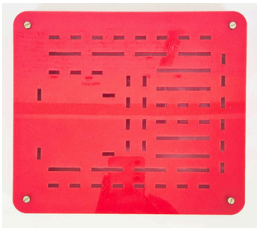
Sequential Lighting and Switching Systems

Session

2

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


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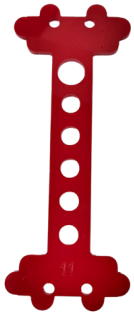


1

Spacers

2

L Clamp

3

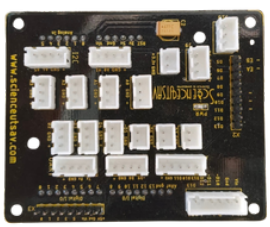
Spacer Screws

4

Nuts

5

Screw

6

Arduino With Shield

7

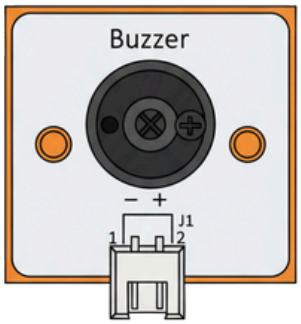
Arduino cable

ELECTRONIC COMPONENTS

DIP LED

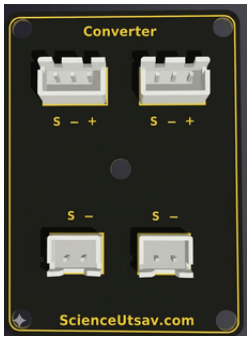
A DIP LED means Dual In-line Package LED — it's the common through-hole type LED you see in electronics projects.

DIP LED is a small light component that glows when electricity flows in one direction.



BUZZER PCB

A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



Converter PCB

The converter PCB is utilized to transform a 3-pin wire into a 2-pin configuration. Since we require 2-pin connections for LEDs & buzzer and our setup only has three direct slots for LEDs and a buzzer, this converter PCB allows us to adapt the 3-pin output into multiple 2-pin outputs. This is particularly useful if we need to connect more than three devices.

2 PIN RMC

Used to connect

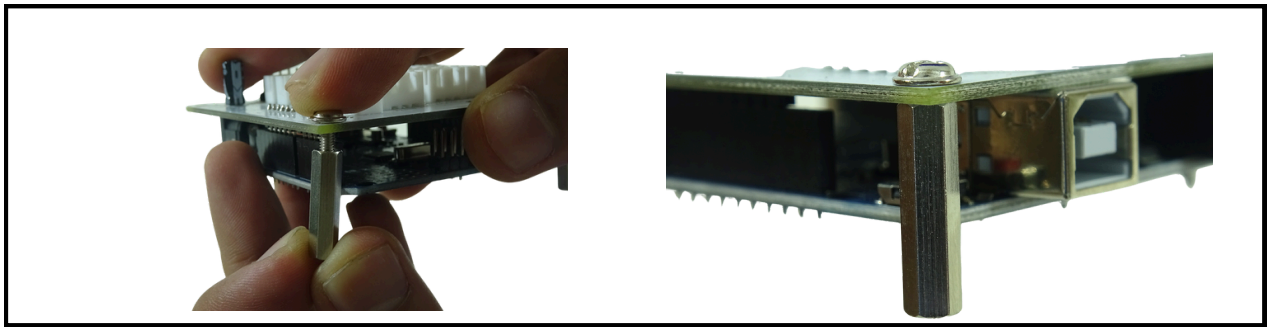
1-watt LEDs

Buzzer.

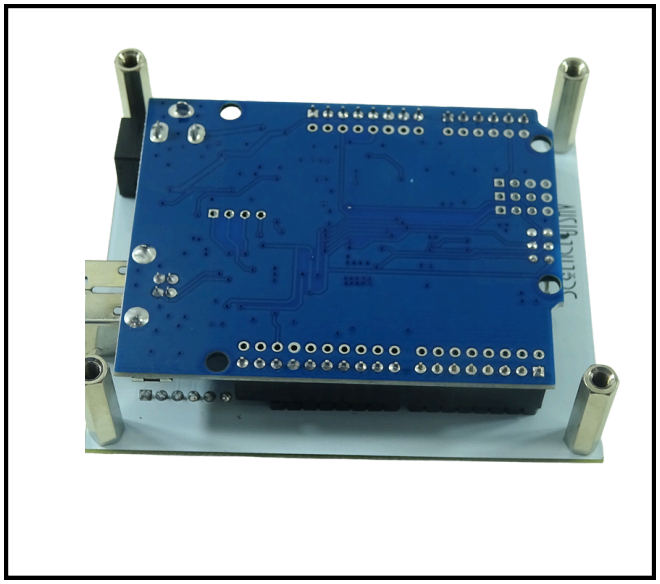
DIP LEDs

Procedure :

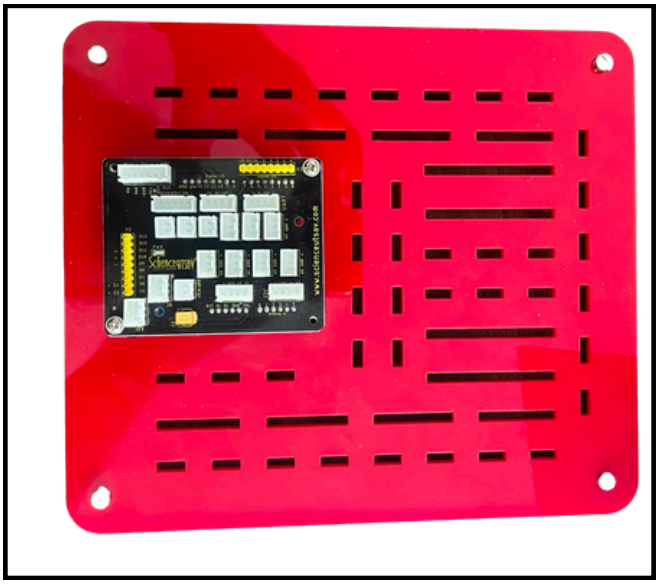
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



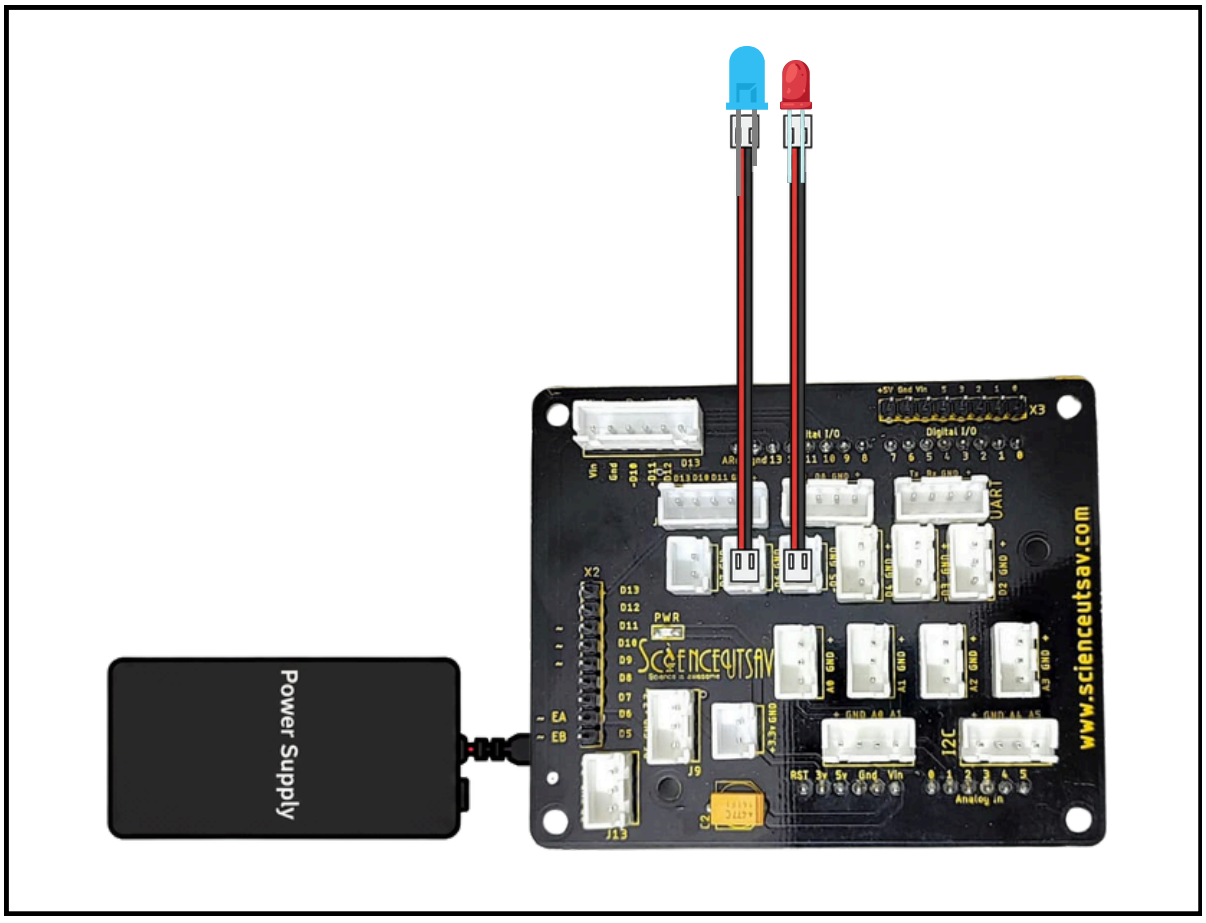
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

The circuit diagram of the project is given below

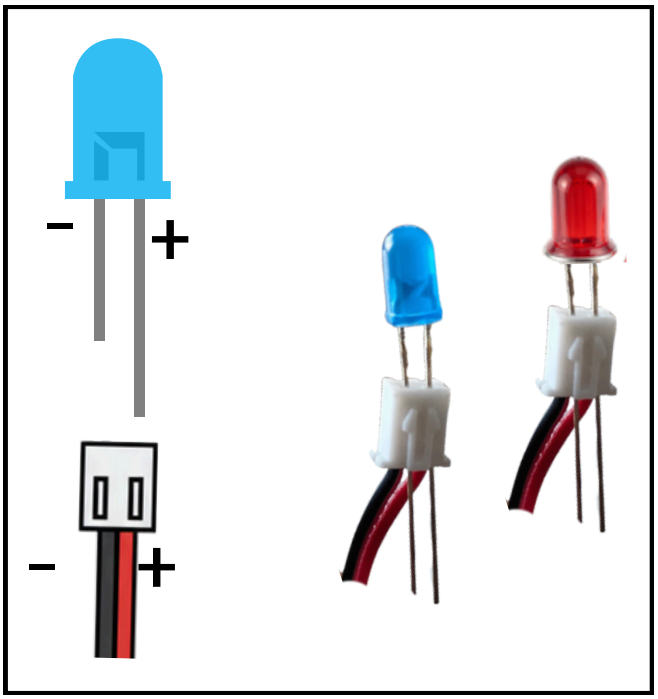


CONNECTION:

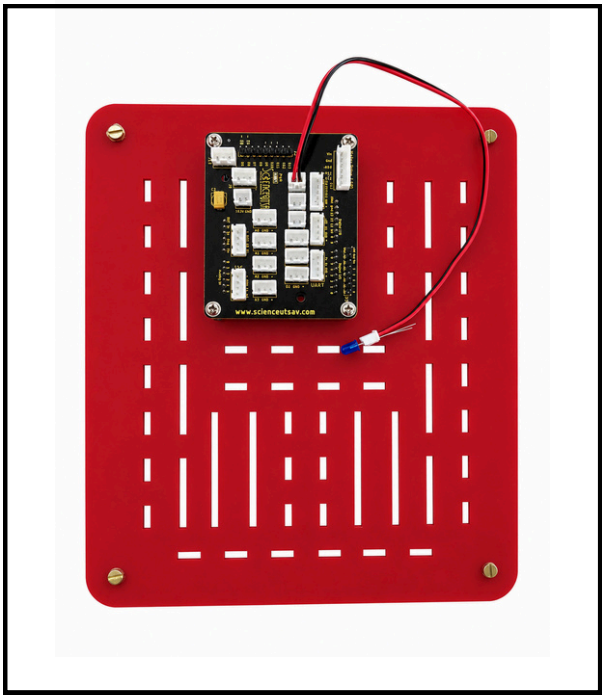
1. RED DIP LED IN D5
2. BLUE DIP LED IN D6

Step 1- Take two 2 Pin RMC cable

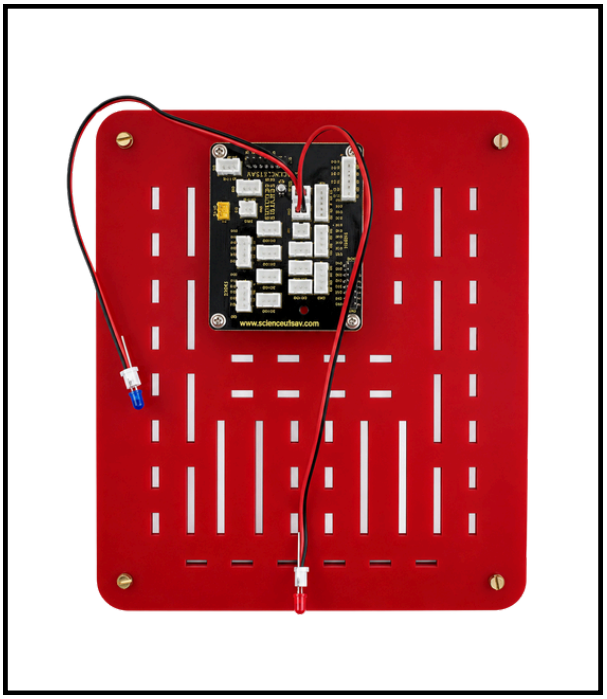
Step 2- Insert the leds from one side , long lead (+)of led to the red wire and short lead(-) of led to black wire



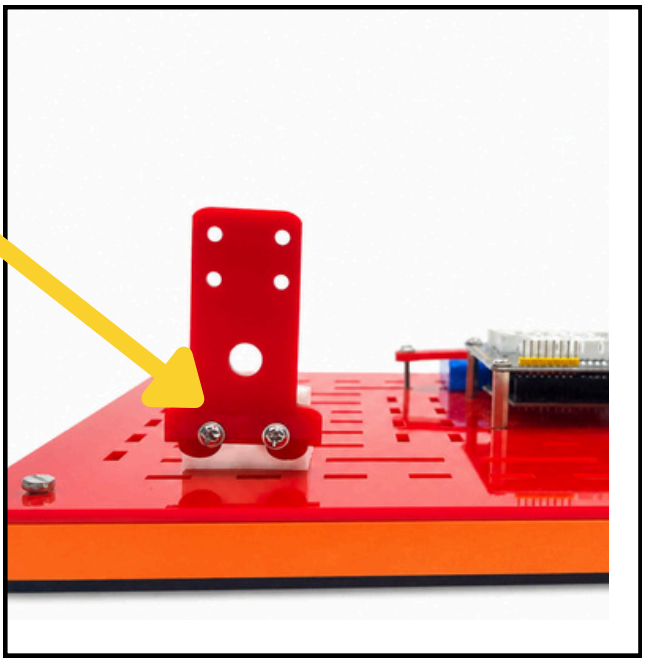
Step 3- Connect the BLUE led to D7



Step 4- Connect the RED led to D6



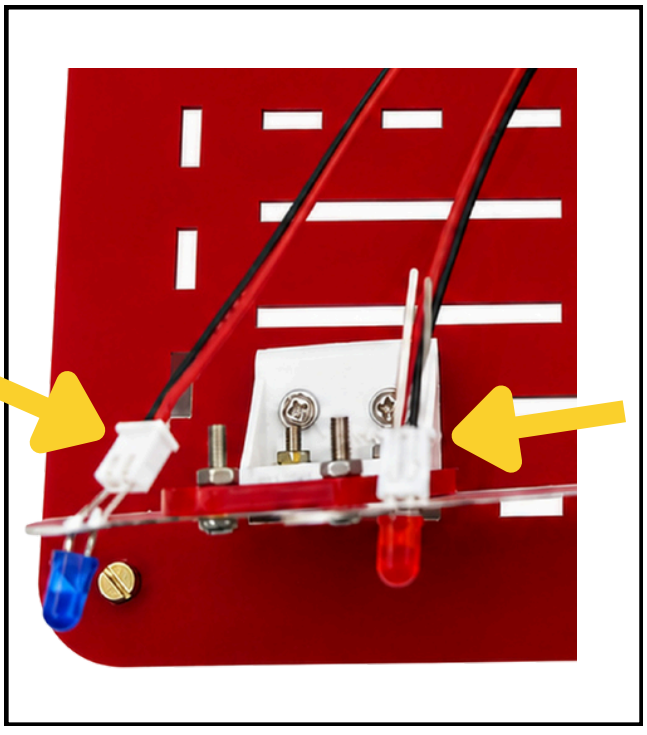
Step 5 - Attach small stand with L- clamp and 3mm nut bolt



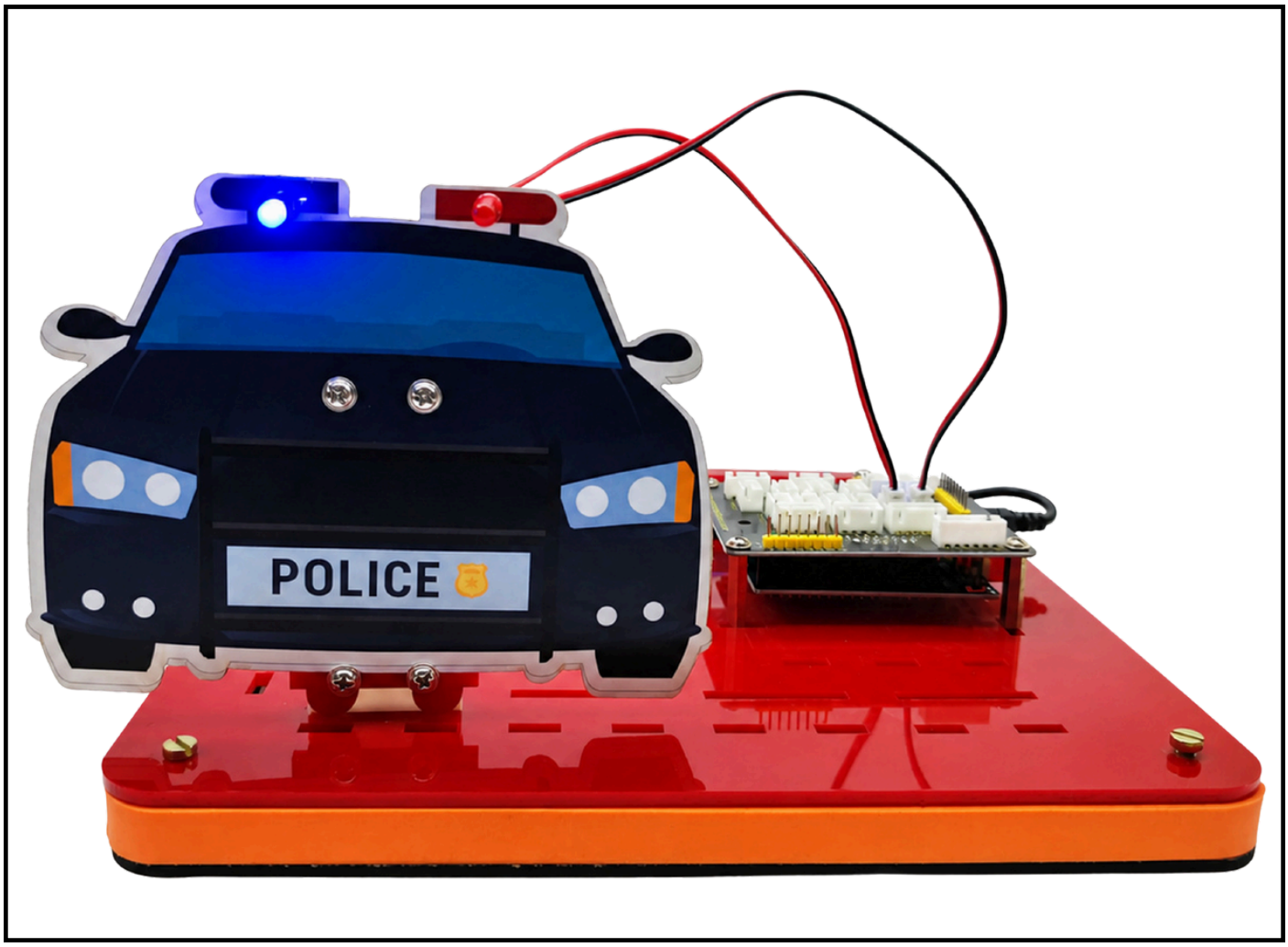
Step 6- Attach car template with 3mm nut and bolt as shown in figure

Step 7- Insert dip led into template

Step 8- Attach 2 pin rmc to LEDs



Upload the code and watch the police light pattern in action.

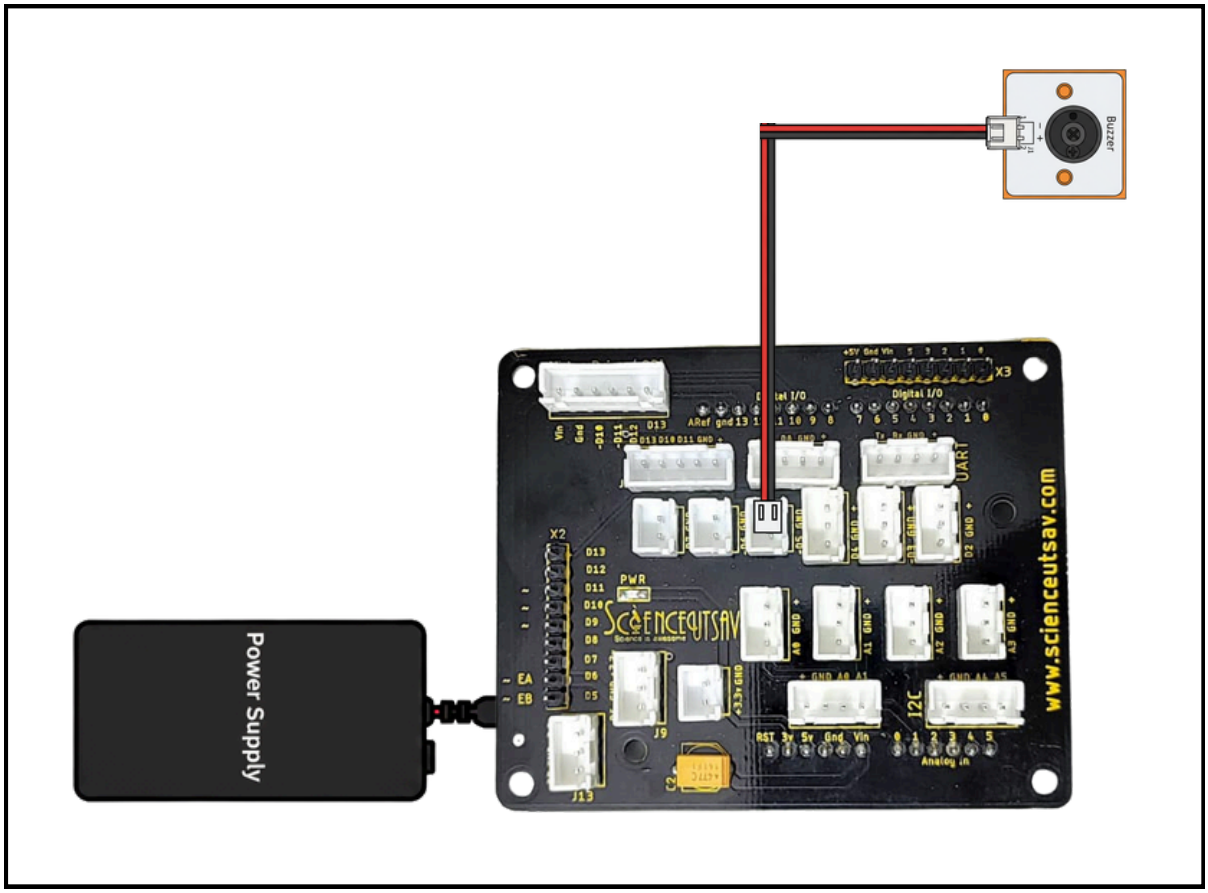


code

```
void setup() {
 pinMode(5, OUTPUT); // redled
 pinMode(6, OUTPUT); // blue led
}

void loop() {
 digitalWrite(5, HIGH);
 digitalWrite(6, LOW);
 delay(200);

 digitalWrite(5, LOW);
 digitalWrite(6, HIGH);
 delay(200);
}
```

BEGINNER PROJECT

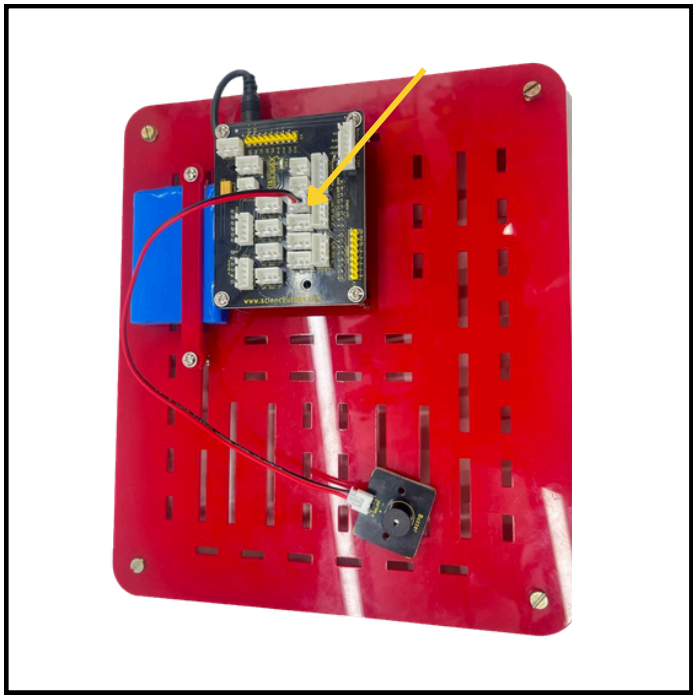
The circuit diagram of the project is given below



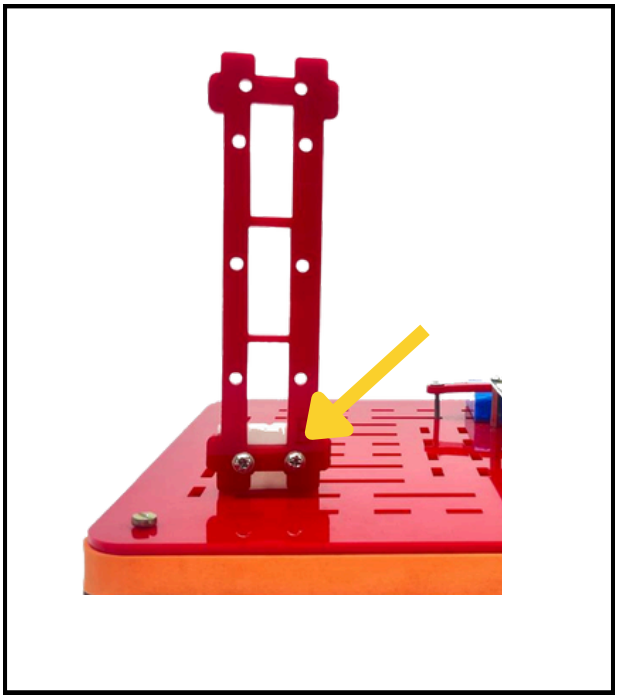
CONNECTION:

1. BUZZER IN D5

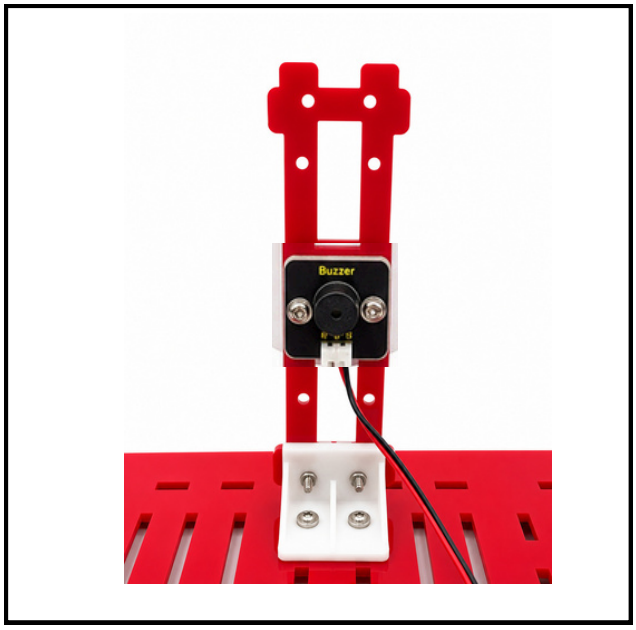
Step 1- Take Buzzer and 2 pin RMC and connect them other end connect to D5 pin in the shield



Step 2- Take multiple led holder and attach it to board with L-clamp and 3mm nut bolt



Step 3- Attach the buzzer to the stand using bolts as shown



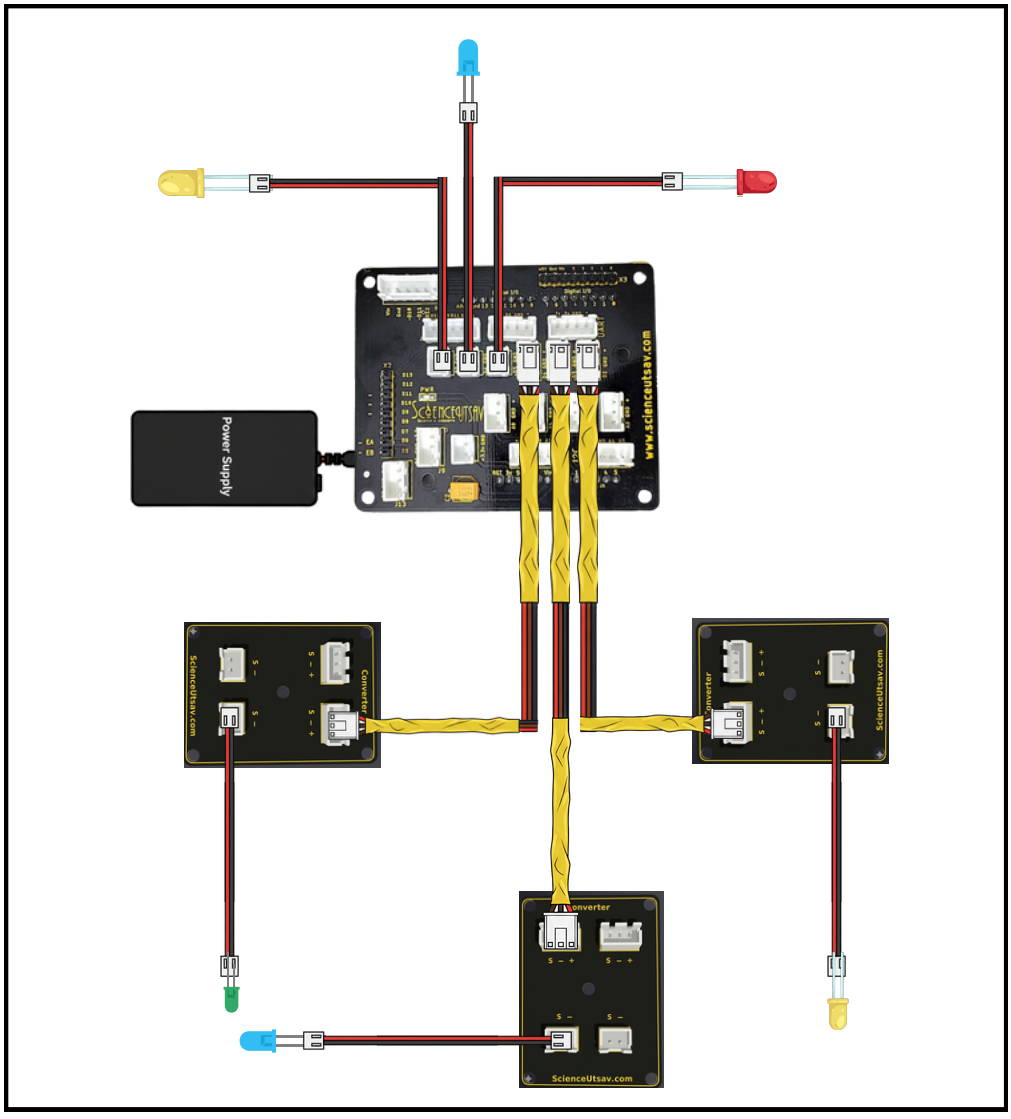
Step 4 - Take ambulance template and attach it to the structure with 3mm nut and bolt

code

```
void setup() {
 pinMode(5, OUTPUT);
}

void loop() {
 digitalWrite(5, HIGH);
 delay(200);

 digitalWrite(5, LOW);
 delay(200);
}
```

INTERMIDIATE PROJECT

The circuit diagram of the project is given below

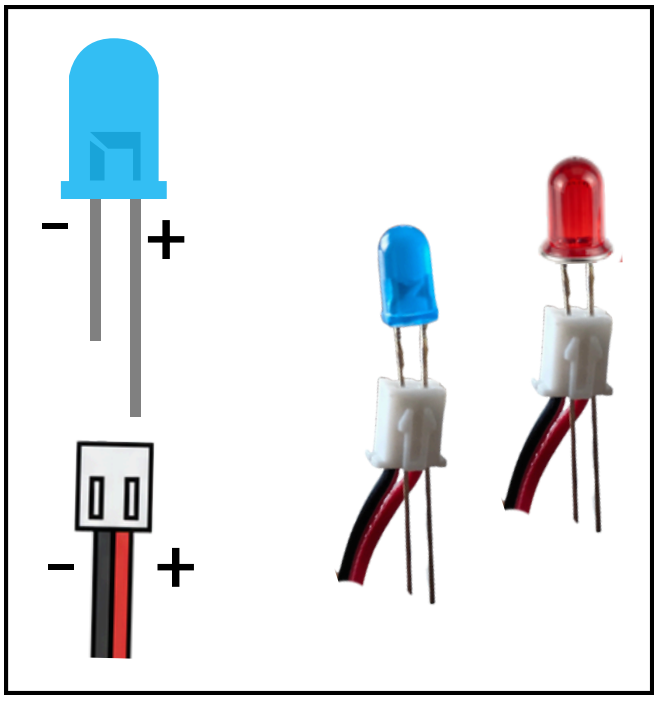


CONNECTION:

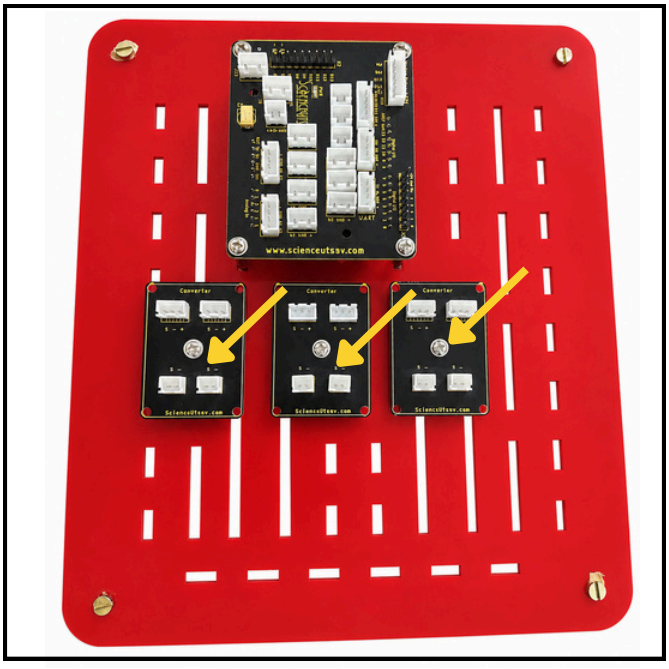
1. LED IN D7
2. LED IN D6
3. LED IN D5
4. CONVERTER PCB IN D4 , CONNECT LED THROUGH CONVERTER PCB
5. CONVERTER PCB IN D3 , CONNECT LED THROUGH CONVERTER PCB
6. CONVERTER PCB IN D2 , CONNECT LED THROUGH CONVERTER PCB

Step1- Take two 2 Pin RMC cable

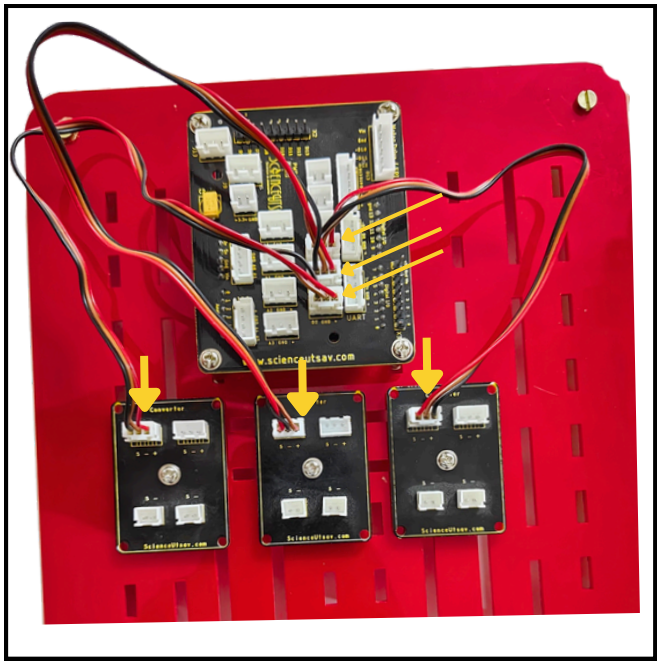
Step 2- Insert the leds from one side , long lead (+)of led to the red wire and short lead(-) of led to black wire



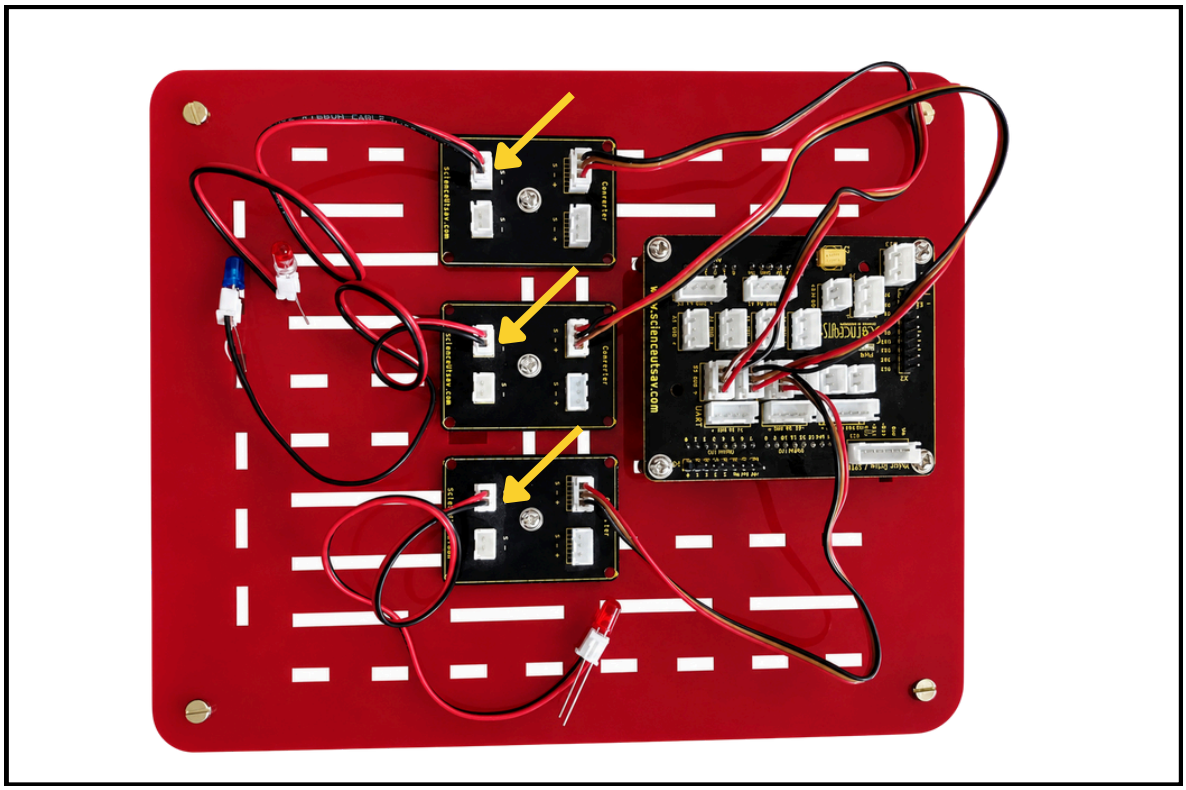
Step 3- Take 3 converter pcb and attach it into board with 3mm nut and boltas shown in figure



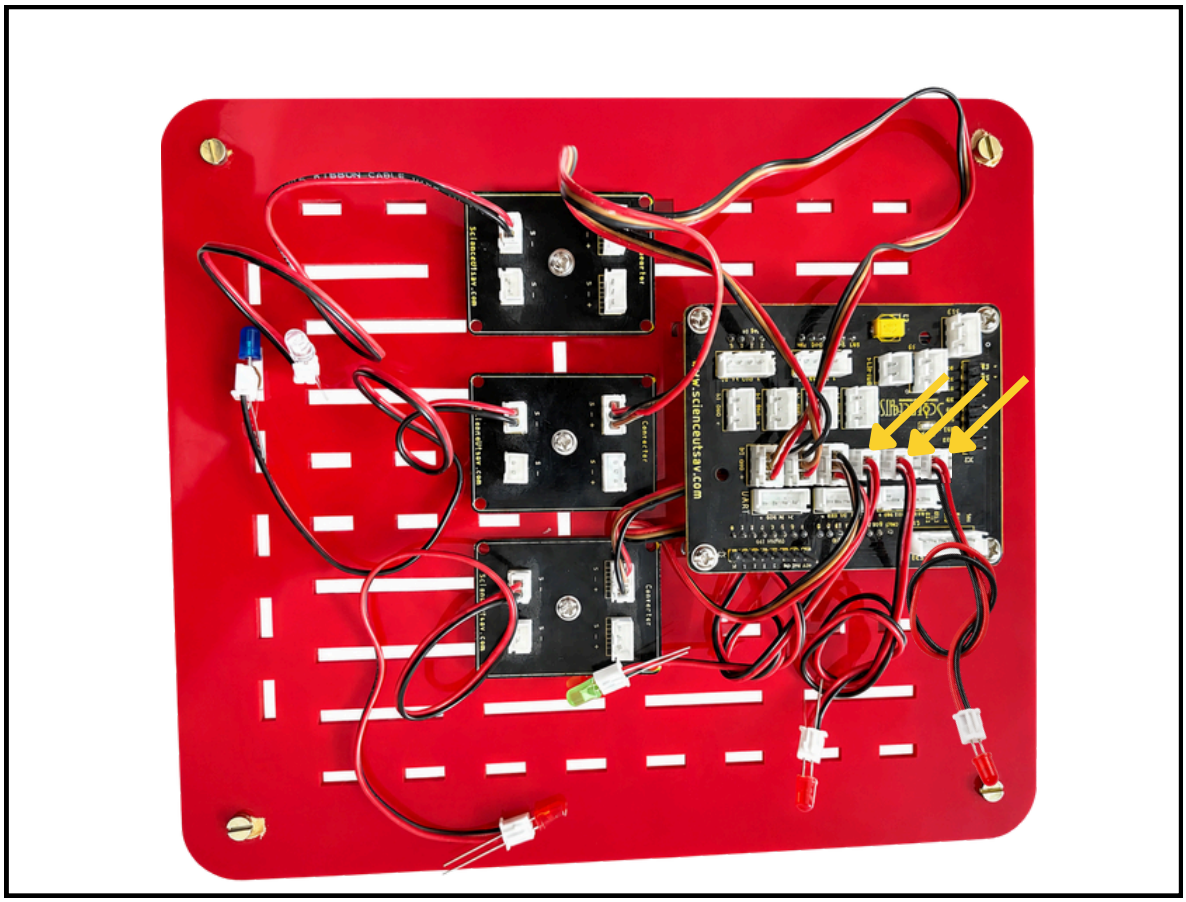
Step 4- Take three 3 pin RMC and connect one end to D2 ,D3,D4 and other end to converter PCBs



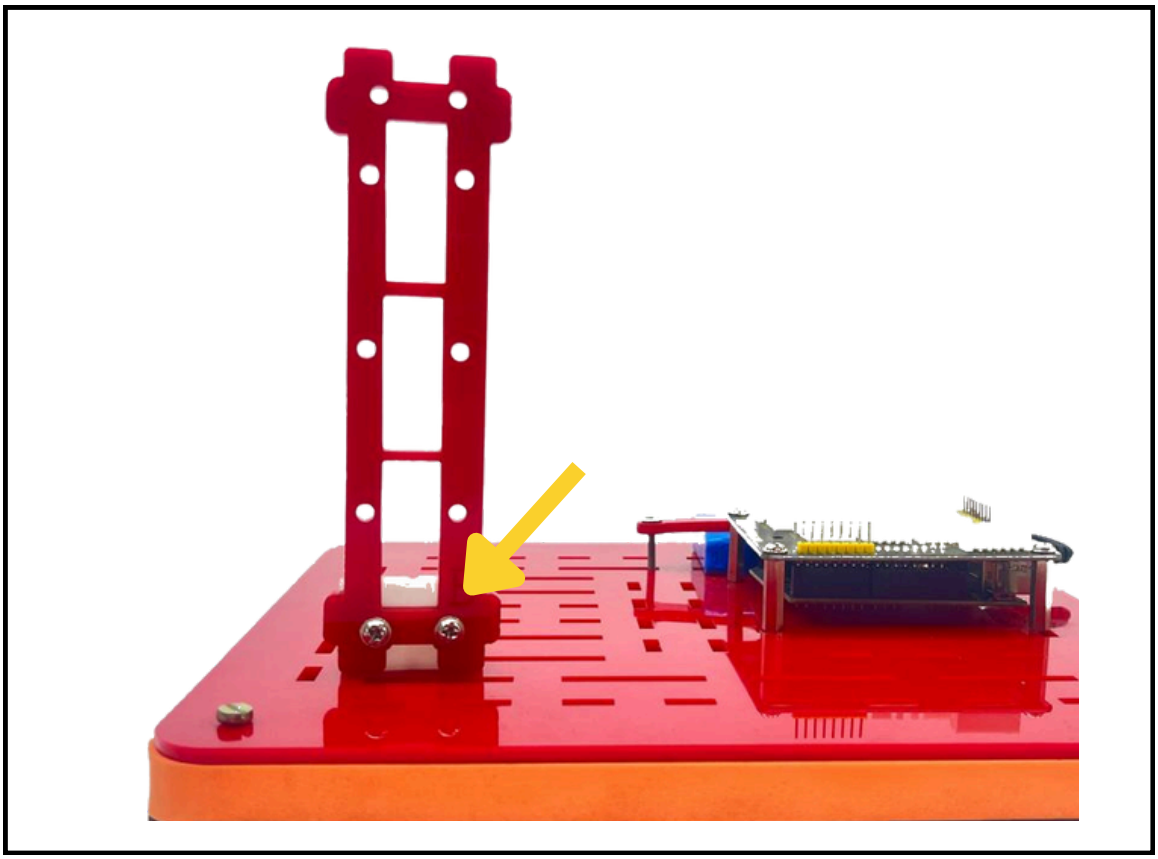
Step 5 - Connect leds to Coverter pcbs 2pin slots



Step 6 - Take three 2 pin RMC and connect it to D5 , D6, D7

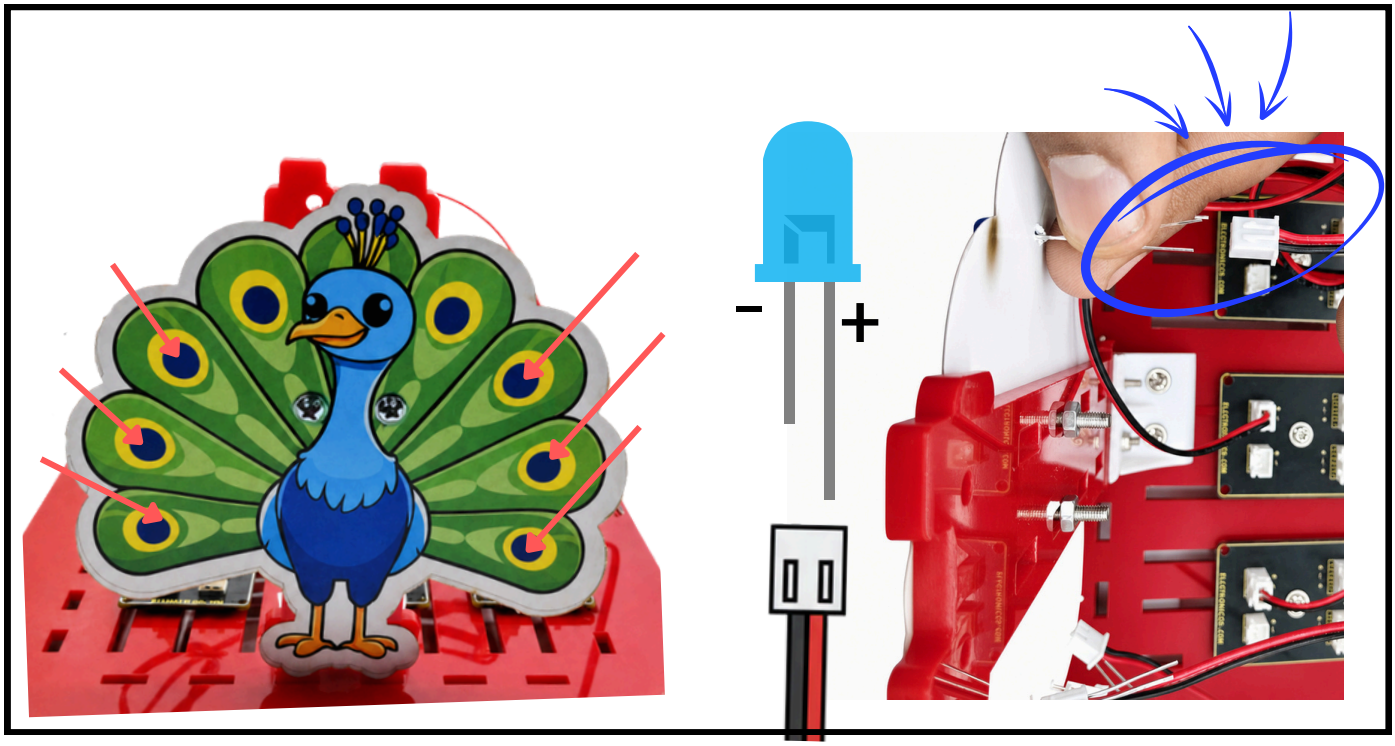


Step 7 - Attach multiple led holder with L-clamp and 3 mm nut & bolt



Step 8 - Attach peacock template to led holder with 3mm nut bolt

Step 9- Insert dip LED from front side of template and connect 2 pin rmc from pin D2 to D6 from back side of template long lead to red and short lead to black



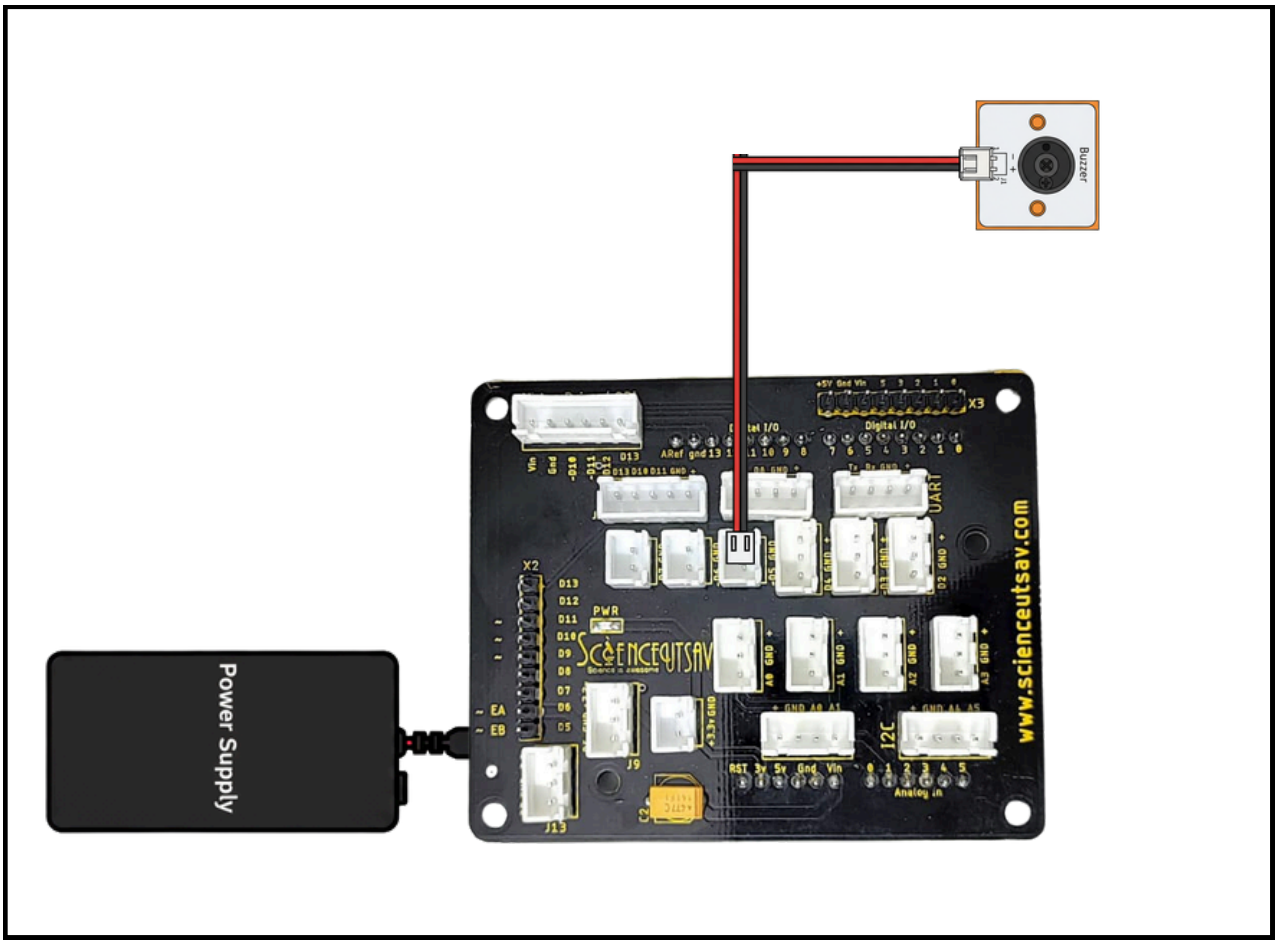
upload the code and observe the led pattern in peocock

code

```
const int dipLed1 = 2;
const int dipLed2 = 3;
const int dipLed3 = 4;
const int dipLed4 = 5;
const int dipLed5 = 6;
const int dipLed6 = 7;
void setup() {
 pinMode(dipLed1, OUTPUT);
 pinMode(dipLed2, OUTPUT);
 pinMode(dipLed3, OUTPUT);
 pinMode(dipLed4, OUTPUT);
 pinMode(dipLed5, OUTPUT);
 pinMode(dipLed6, OUTPUT);
}
void loop() {
 // Pair 1: Pin 7 and Pin 2 ON together
 digitalWrite(5, HIGH);
 digitalWrite(2, HIGH);
 delay(300);
 digitalWrite(2, LOW);
 digitalWrite(5, LOW);
 // Pair 2: Pin 6 and Pin 3 ON together
 digitalWrite(6, HIGH);
 digitalWrite(3, HIGH);
 delay(300);
 digitalWrite(6, LOW);
 digitalWrite(3, LOW);
 // Pair 3: Pin 4 and Pin 5 ON together
 digitalWrite(4, HIGH);
 digitalWrite(7, HIGH);
 delay(300);
 digitalWrite(4, LOW);
 digitalWrite(7, LOW);
 delay(300);
}
```

INTERMIDIATE PROJECT

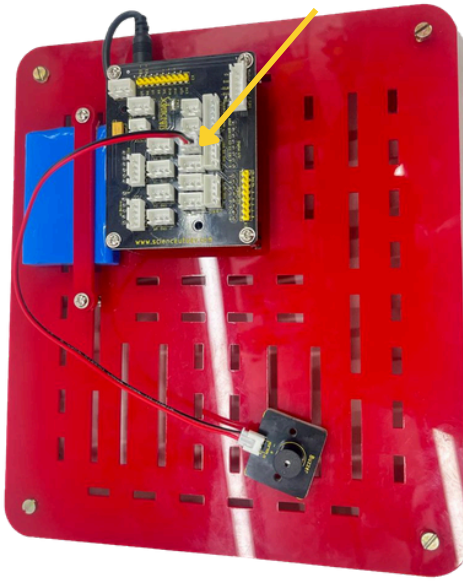
The circuit diagram of the project is given below



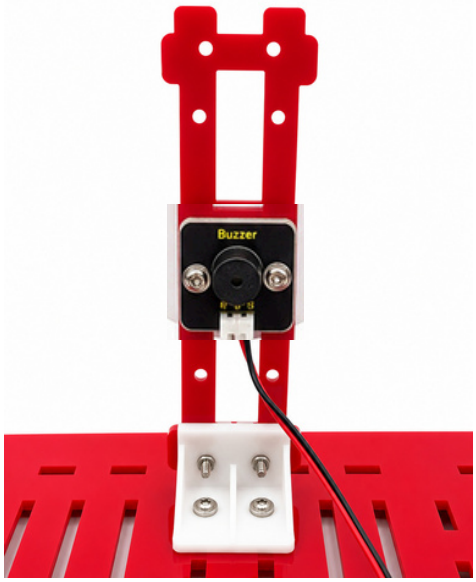
CONNECTION:

1. BUZZER IN D5

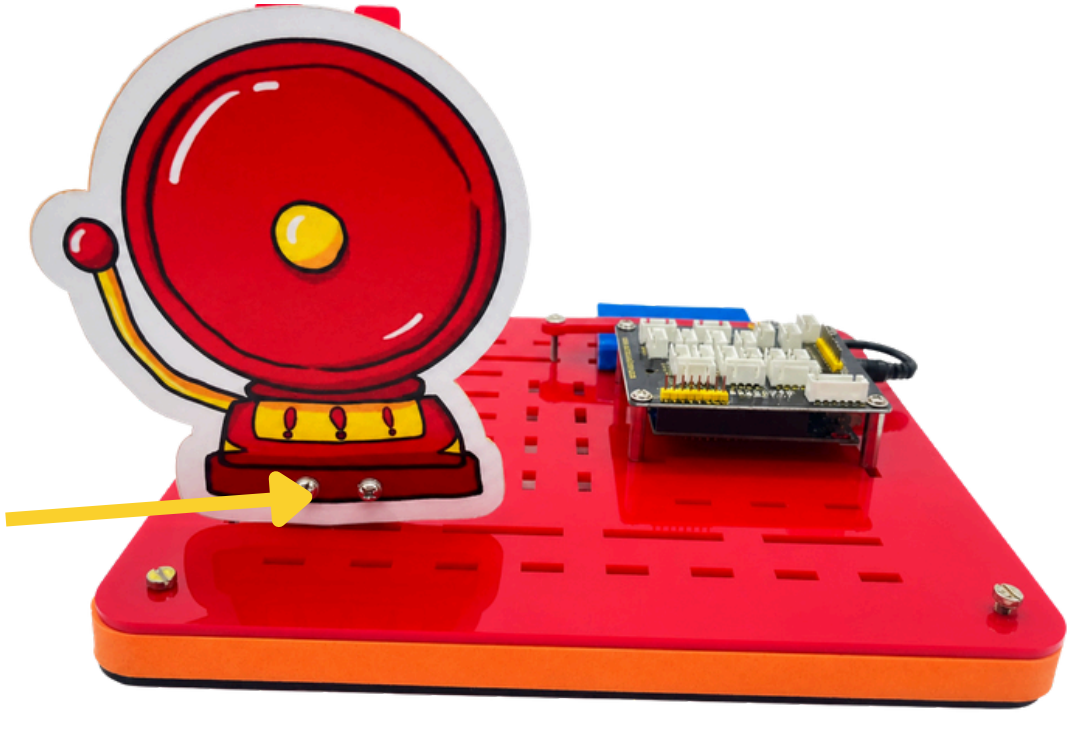
Step 1- Take Buzzer and 2 pin RMC and connect them other end connect to D5 pin in the shield



Step 2- Fix the buzzer to the stand as shown.



Step 3 - Attach the template to the stand as shown

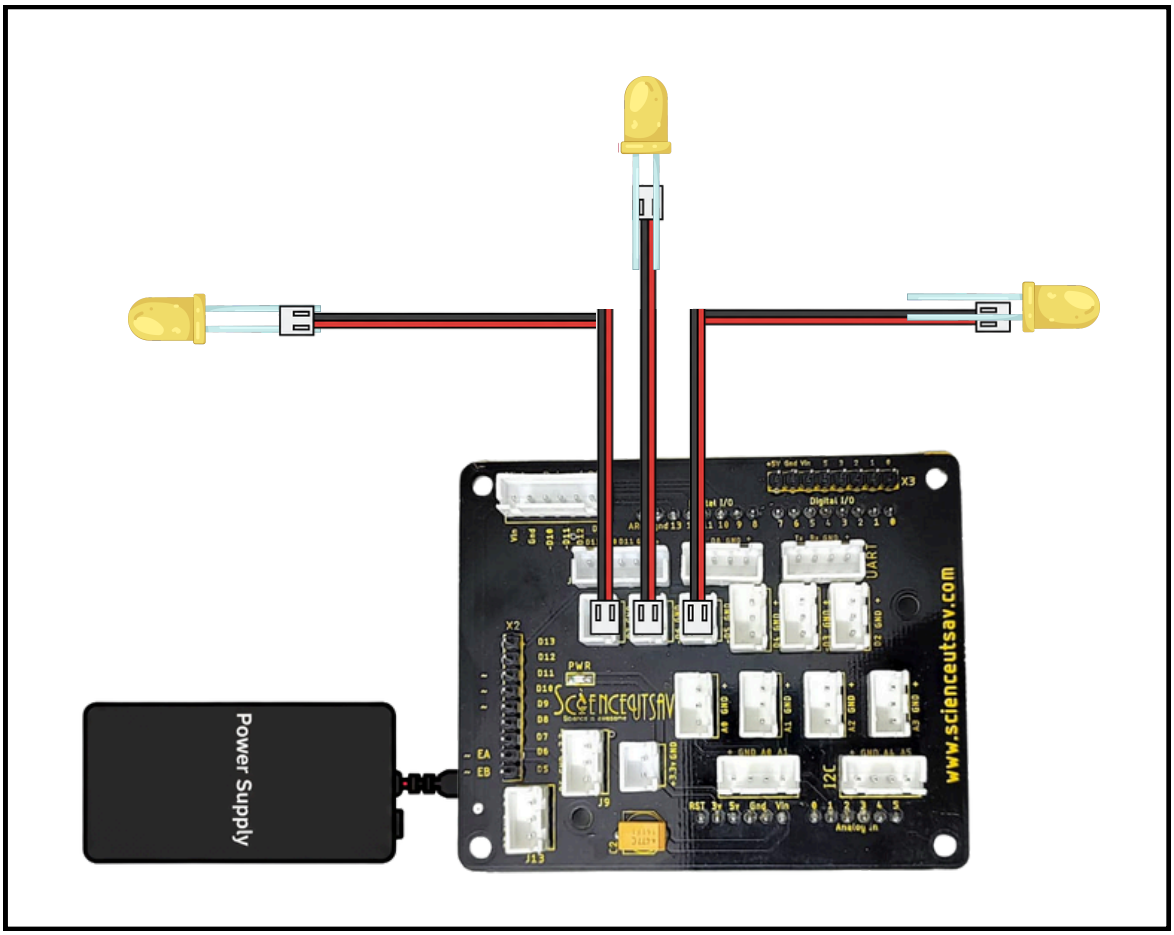


code

```
void setup() {
 pinMode(5, OUTPUT);
}

void loop() {
 digitalWrite(5, HIGH);
 delay(200);

 digitalWrite(5, LOW);
 delay(200);
}
```

ADVANCE PROJECT

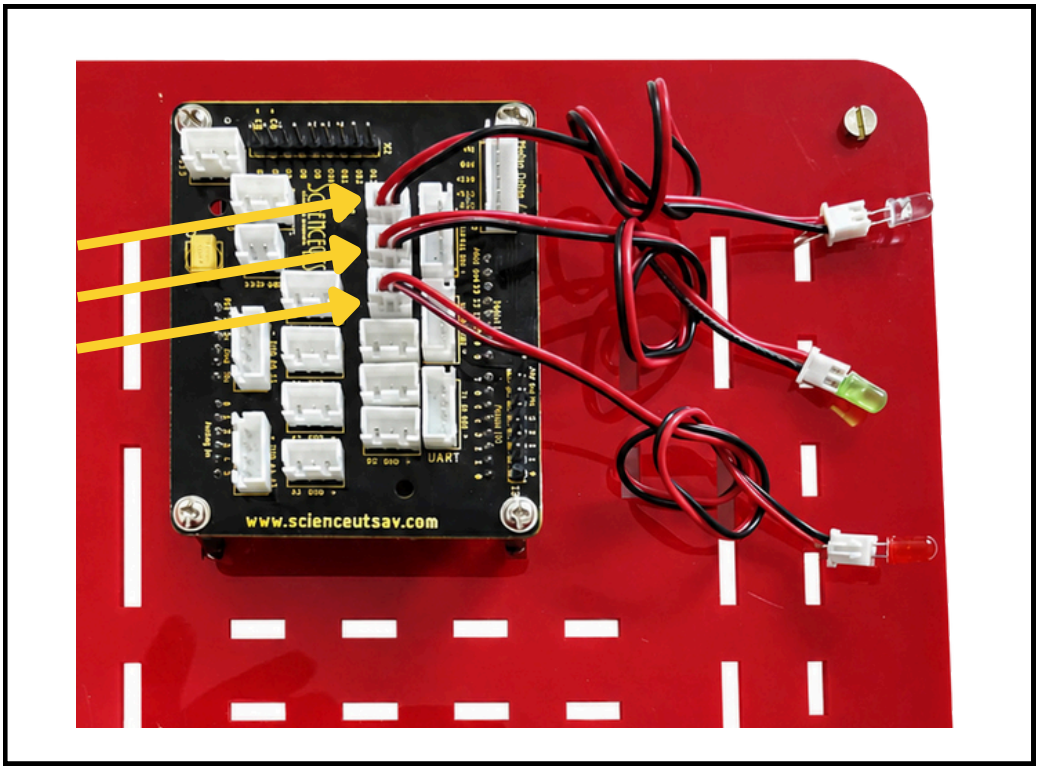
The circuit diagram of the project is given below



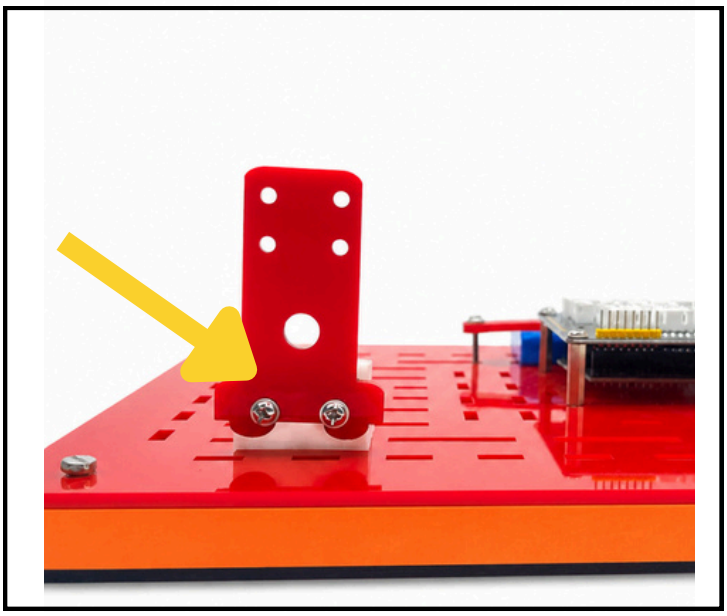
CONNECTION:

1. DIP LED IN D5
2. DIP LED IN D6
3. DIP LED IN D7

Step 1- Take three 2 Pin RMC cable, connect it to D5 , D6, D7 pin in the shield



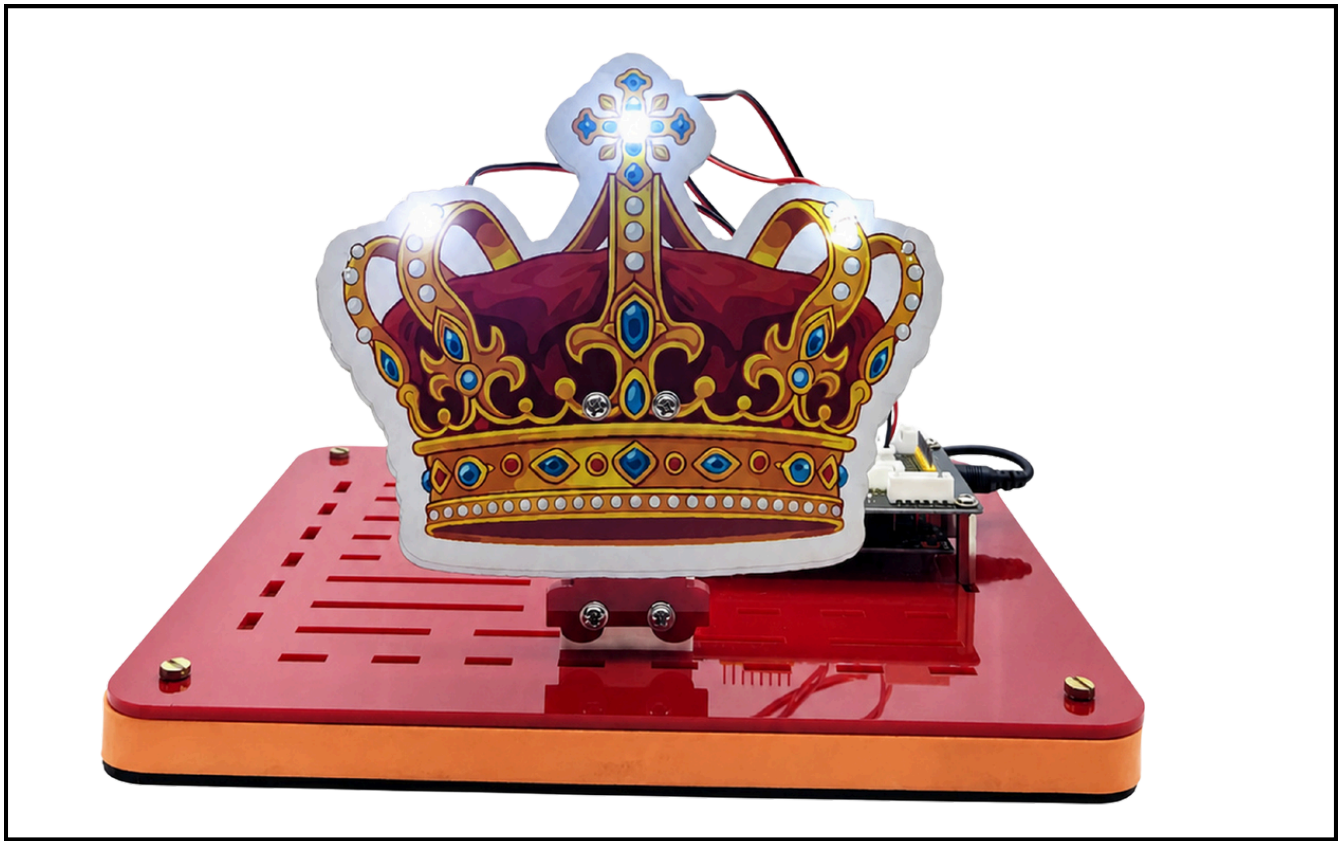
Step 2 - Attach small stand using L- clamp and 3mm nut and bolt



Step 3 - Attach crown with 3mm nut and bolt

Step 4: Take three DIP LEDs and insert them into the template front side and connect the 2 pin rmc to leds from back side of template

upload code and observe the led pattern in crown

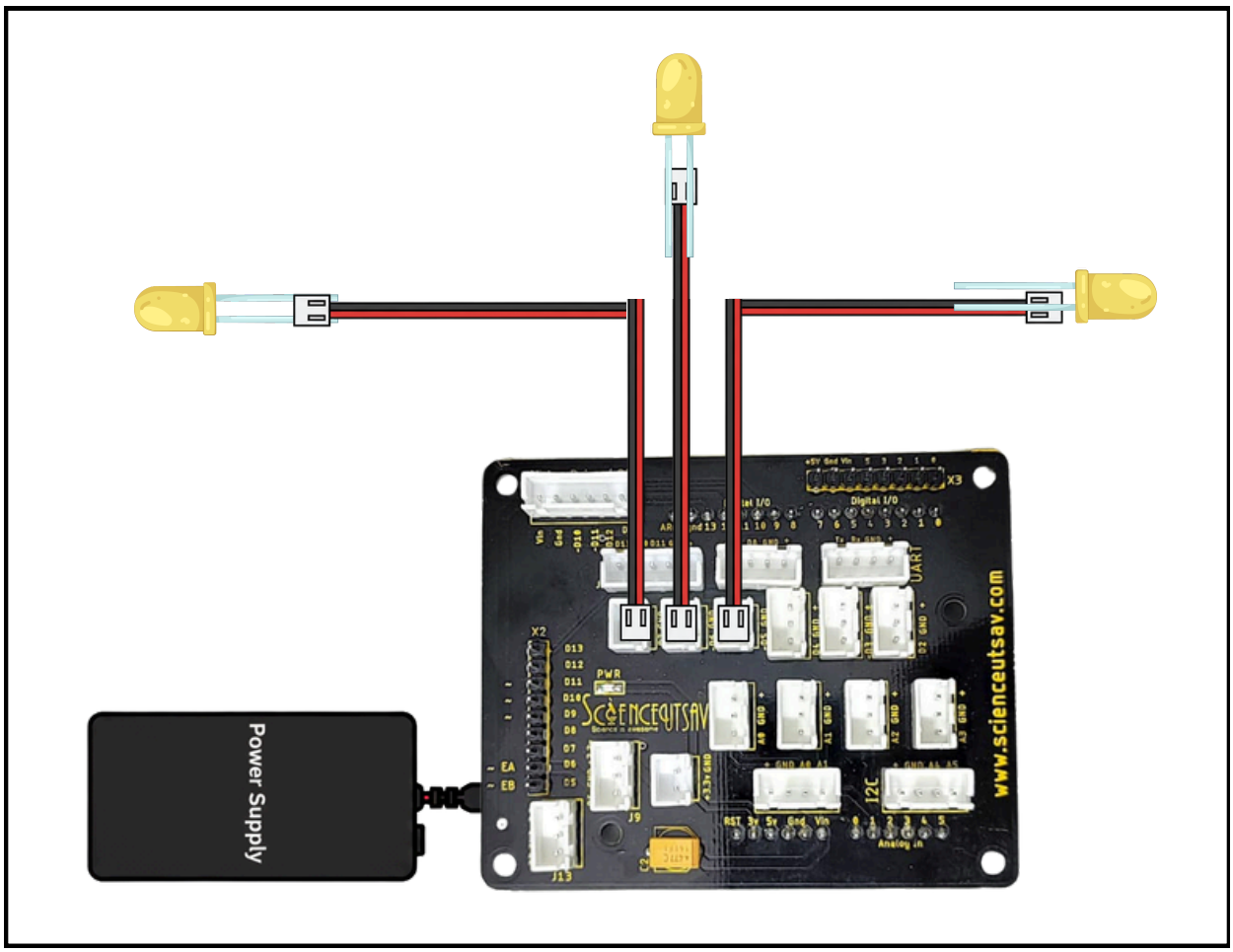


code

```
const int dipLed1 = 5;
const int dipLed2 = 6;
const int dipLed3 = 7;

void setup() {
 pinMode(dipLed1, OUTPUT);
 pinMode(dipLed2, OUTPUT);
 pinMode(dipLed3, OUTPUT);
}
void loop() {
 digitalWrite(dipLed1, HIGH);
 digitalWrite(dipLed2, HIGH);
 digitalWrite(dipLed3, LOW);
 delay(400);
 digitalWrite(dipLed1, LOW);
 digitalWrite(dipLed2, LOW);
 digitalWrite(dipLed3, HIGH);
 delay(400);
}
```

ADVANCE PROJECT

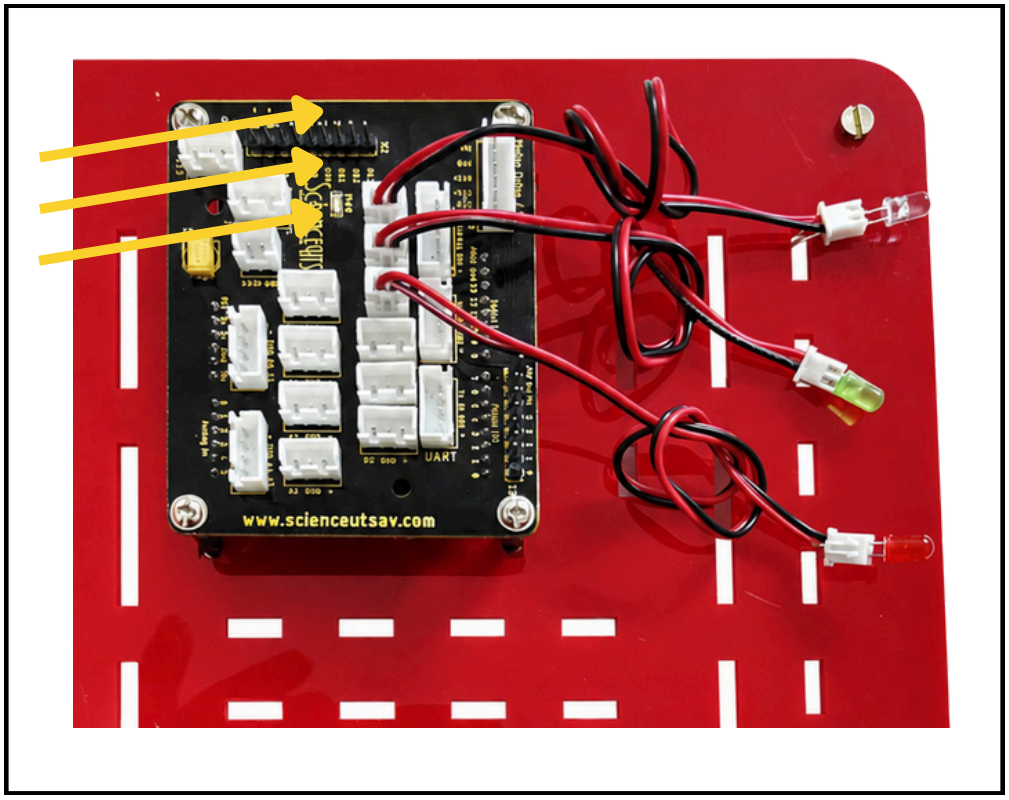
The circuit diagram of the project is given below



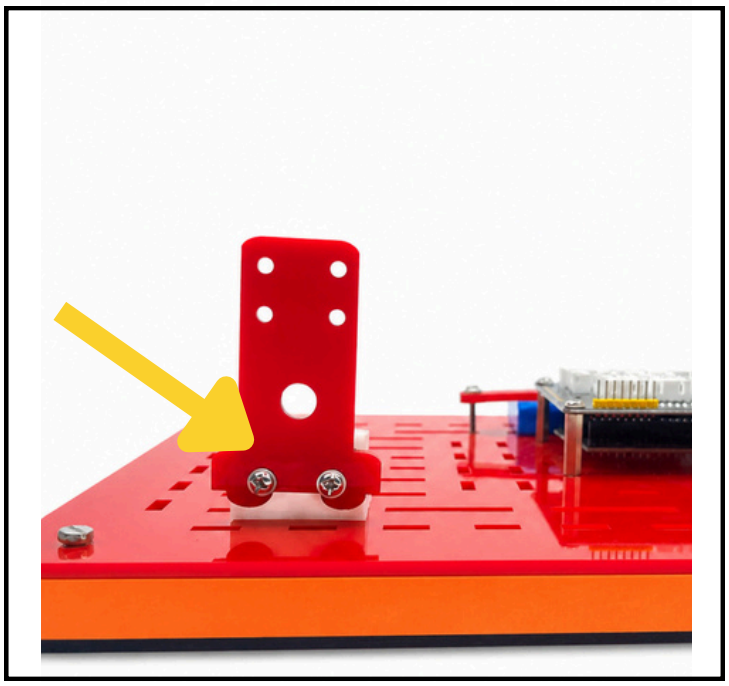
CONNECTION:

1. DIP LED IN D5
2. DIP LED IN D6
3. DIP LED IN D7

Step 1- Take three 2 Pin RMC cable, connect it to D5 , D6, D7 pin in the shield



Step 2 - Attach small stand using L- clamp and 3mm nut and bolt



Step 3 - Attach template with 3mm nut and bolt

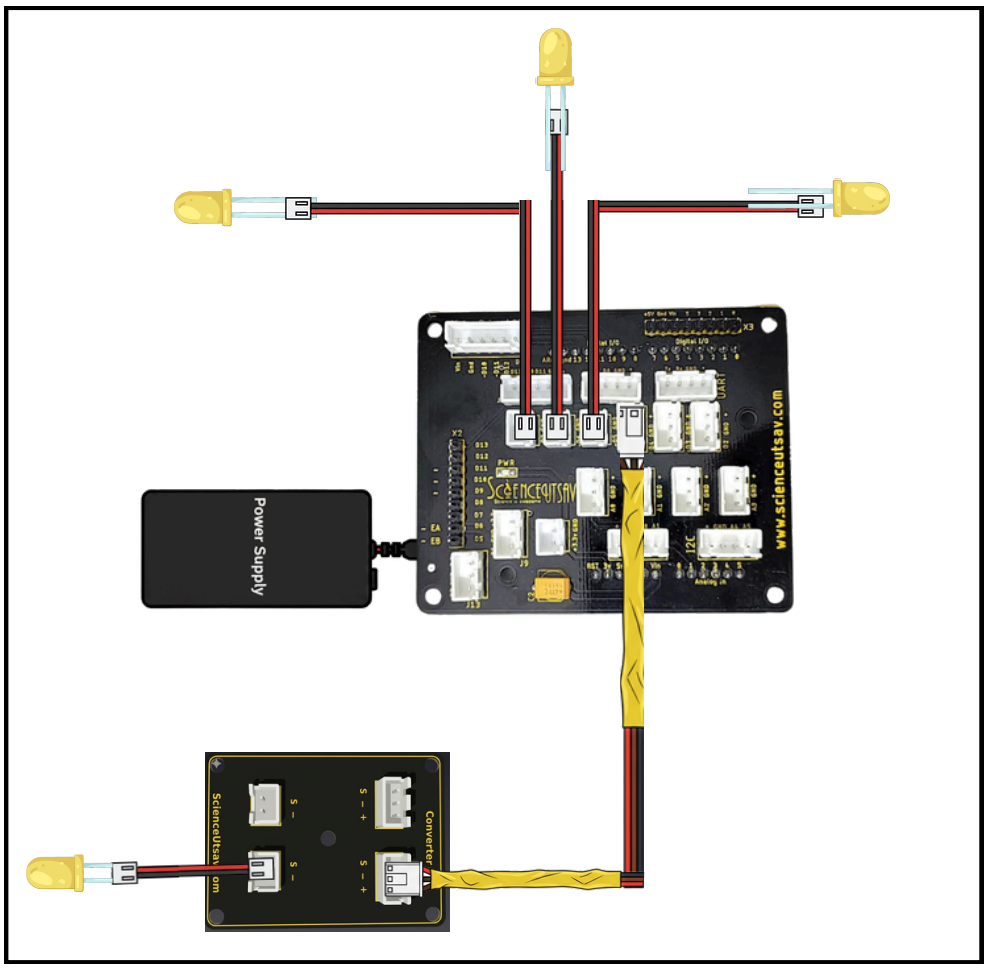
upload the code and observe the pattern in template

code

```
const int dipLed1 = 5;
const int dipLed2 = 6;
const int dipLed3 = 7;
void setup() {
 pinMode(dipLed1, OUTPUT);
 pinMode(dipLed2, OUTPUT);
 pinMode(dipLed3, OUTPUT);

}
void loop() {
 digitalWrite(dipLed1, HIGH);
 digitalWrite(dipLed2, HIGH);
 digitalWrite(dipLed3, LOW);
 delay(400);
 digitalWrite(dipLed1, LOW);
 digitalWrite(dipLed2, LOW);
 digitalWrite(dipLed3, HIGH);
 delay(400);
}
```

EXPERT PROJECT

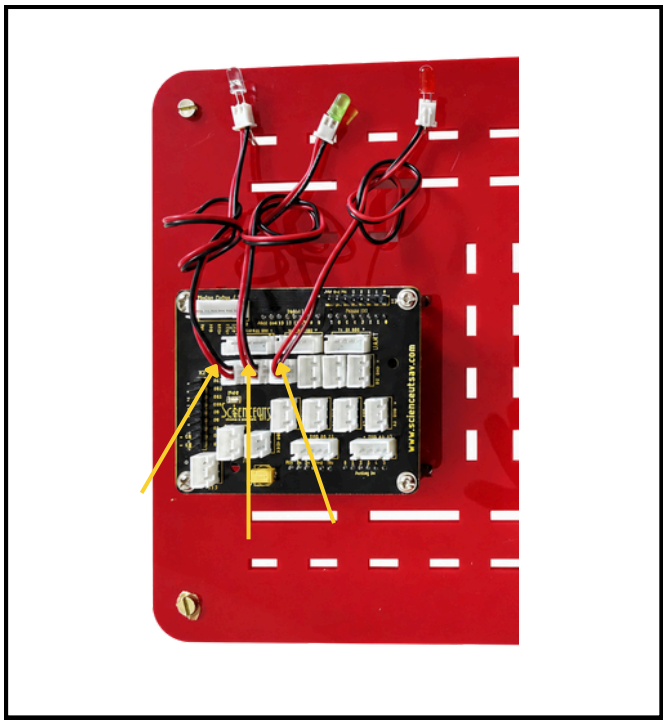
The circuit diagram of the project is given below



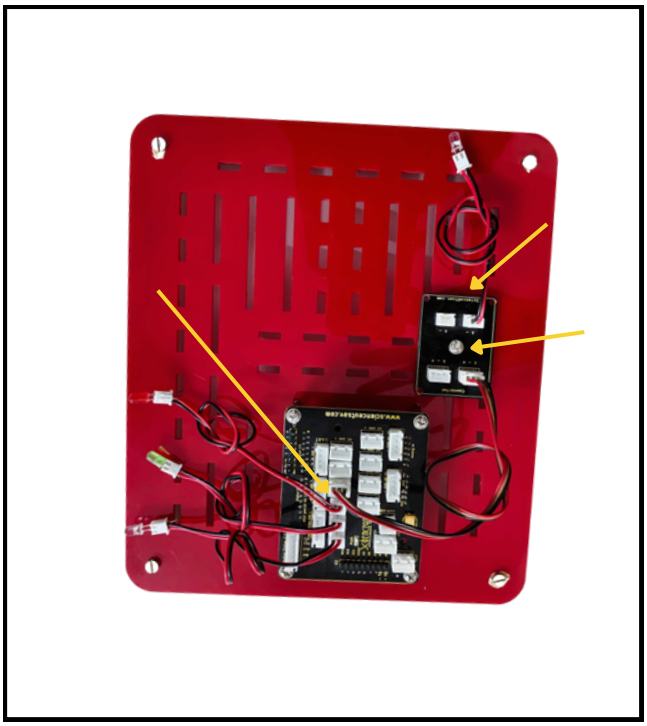
CONNECTION:

1. DIP LED IN D5
2. DIP LED IN D6
3. DIP LED IN D7
4. CONVERTER PCB IN D4, LED TO CONVERTER PCB

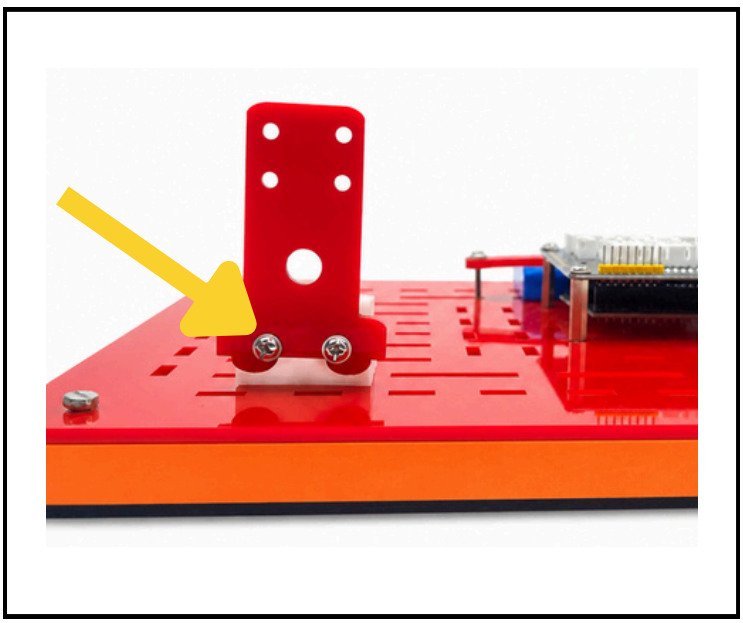
Step 1- Take three 2 Pin RMC cable connect it to d5 d6 d7



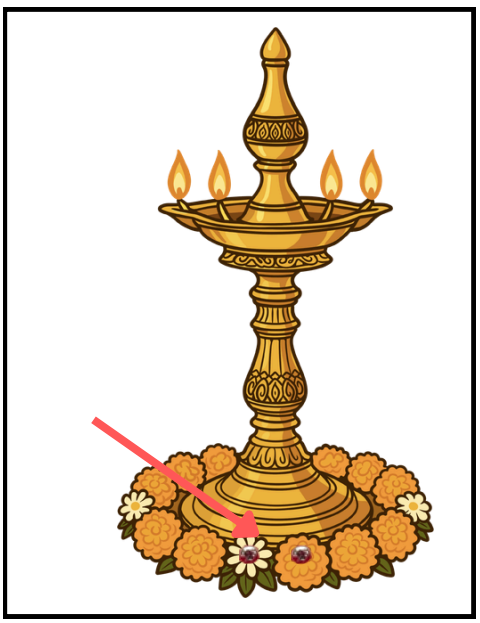
Step 2- Attach converter pcb to board and take 3 pin rmc connect one end to d4 pin in the shield



Step 3 - Attach small stand using L- clamp and 3mm nut and bolt



Step 4 - Attach template with 3mm nut and bolt



upload and observe led pattern in template

code

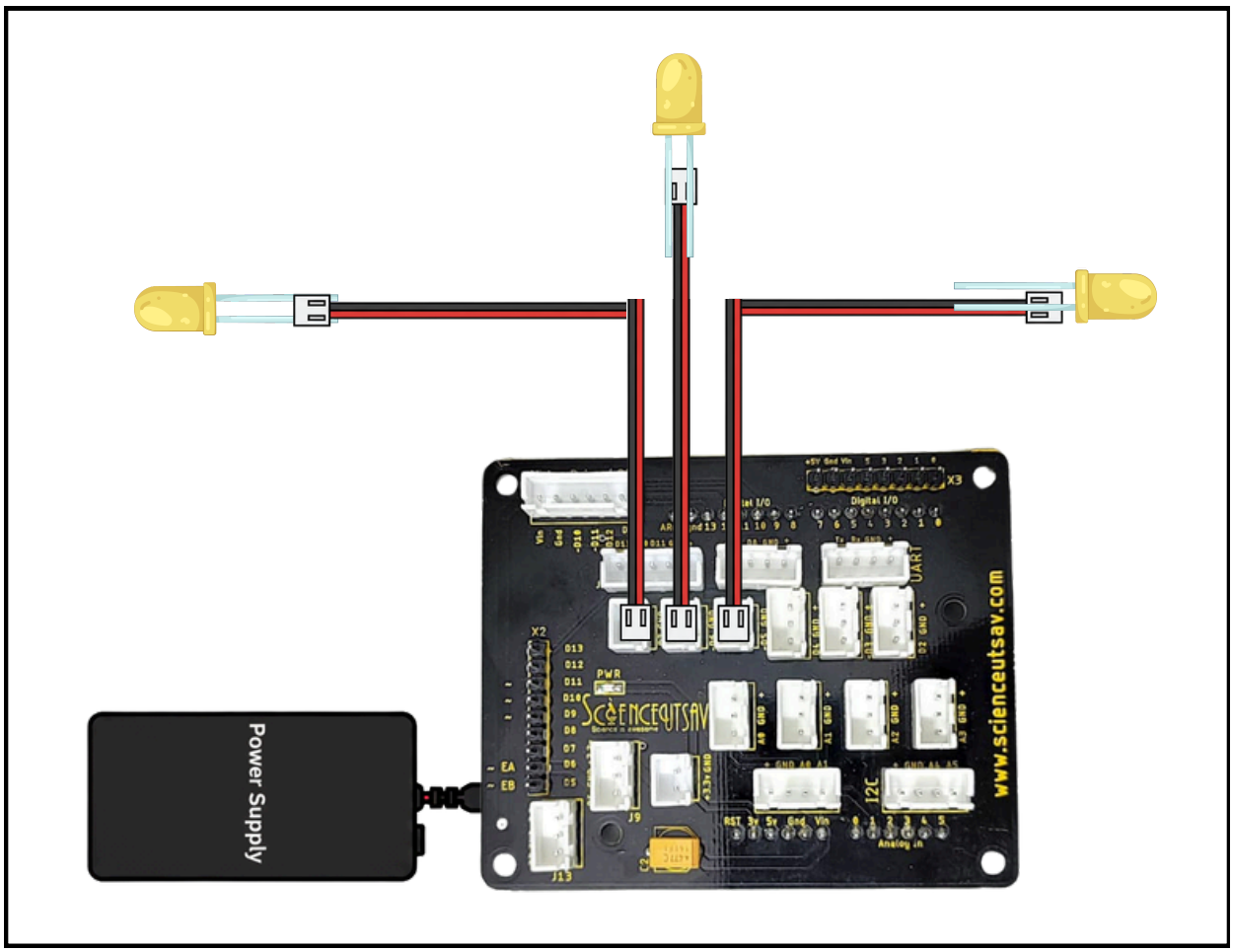
```
const int dipLed1 = 4;
const int dipLed2 = 5;
const int dipLed3 = 6;
const int dipLed4 = 7;

void setup() {
 pinMode(dipLed1, OUTPUT);
 pinMode(dipLed2, OUTPUT);
 pinMode(dipLed3, OUTPUT);
 pinMode(dipLed4, OUTPUT);
}
void loop() {
 // All 4 LEDs ON together
 digitalWrite(dipLed1, HIGH);
 digitalWrite(dipLed2, HIGH);
 digitalWrite(dipLed3, HIGH);
 digitalWrite(dipLed4, HIGH);
 delay(500);
 // OFF one by one
```

code

```
 digitalWrite(dipLed1, LOW);
 delay(300);

 digitalWrite(dipLed2, LOW);
 delay(300);

 digitalWrite(dipLed3, LOW);
 delay(300);

 digitalWrite(dipLed4, LOW);
 delay(500);
}
```

EXPERT PROJECT

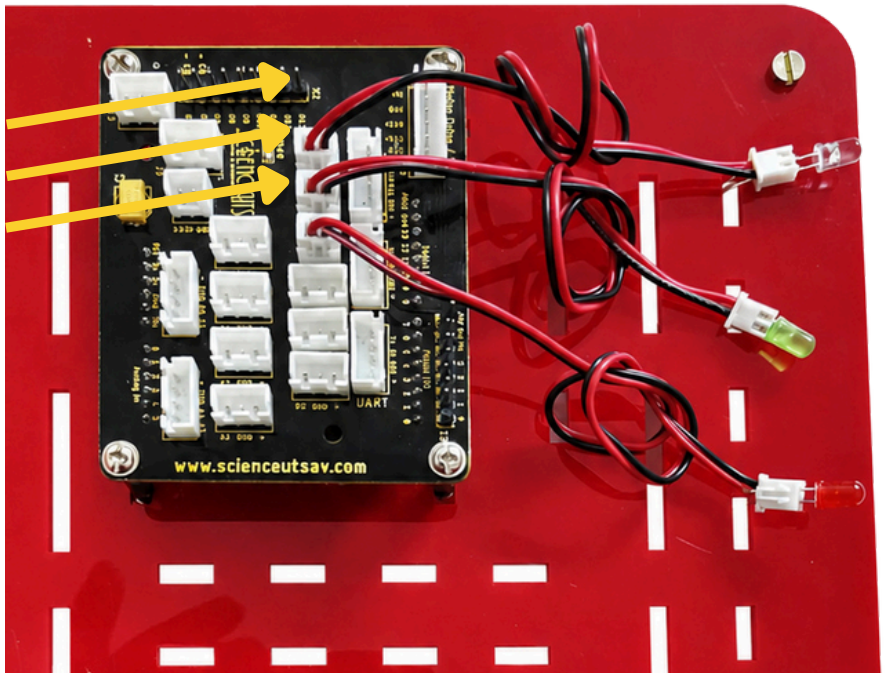
The circuit diagram of the project is given below



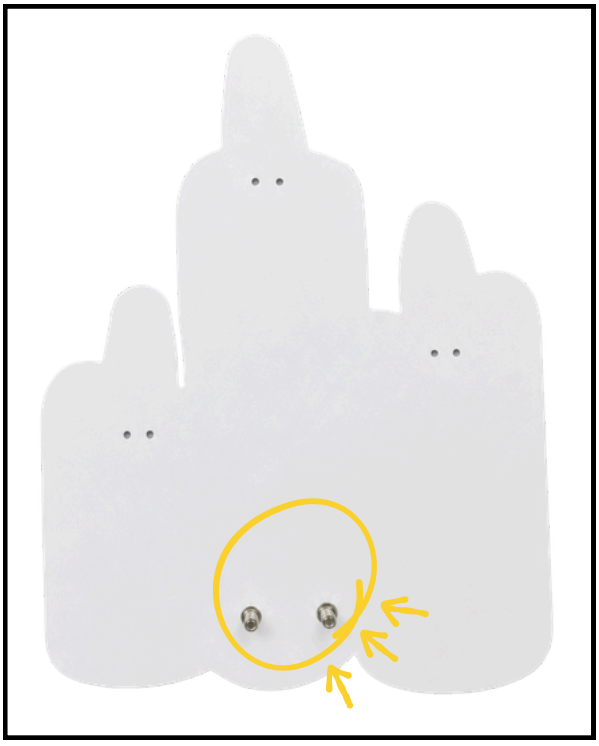
CONNECTION:

1. DIP LED IN D5
2. DIP LED IN D6
3. DIP LED IN D7

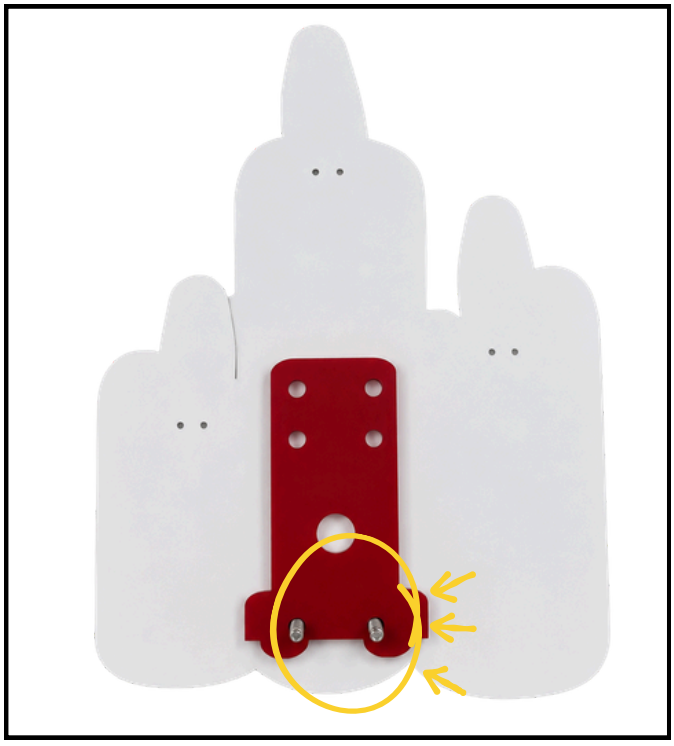
Step1- Take three 2 Pin RMC cable, connect it to D5 , D6, D7 pin in the shield



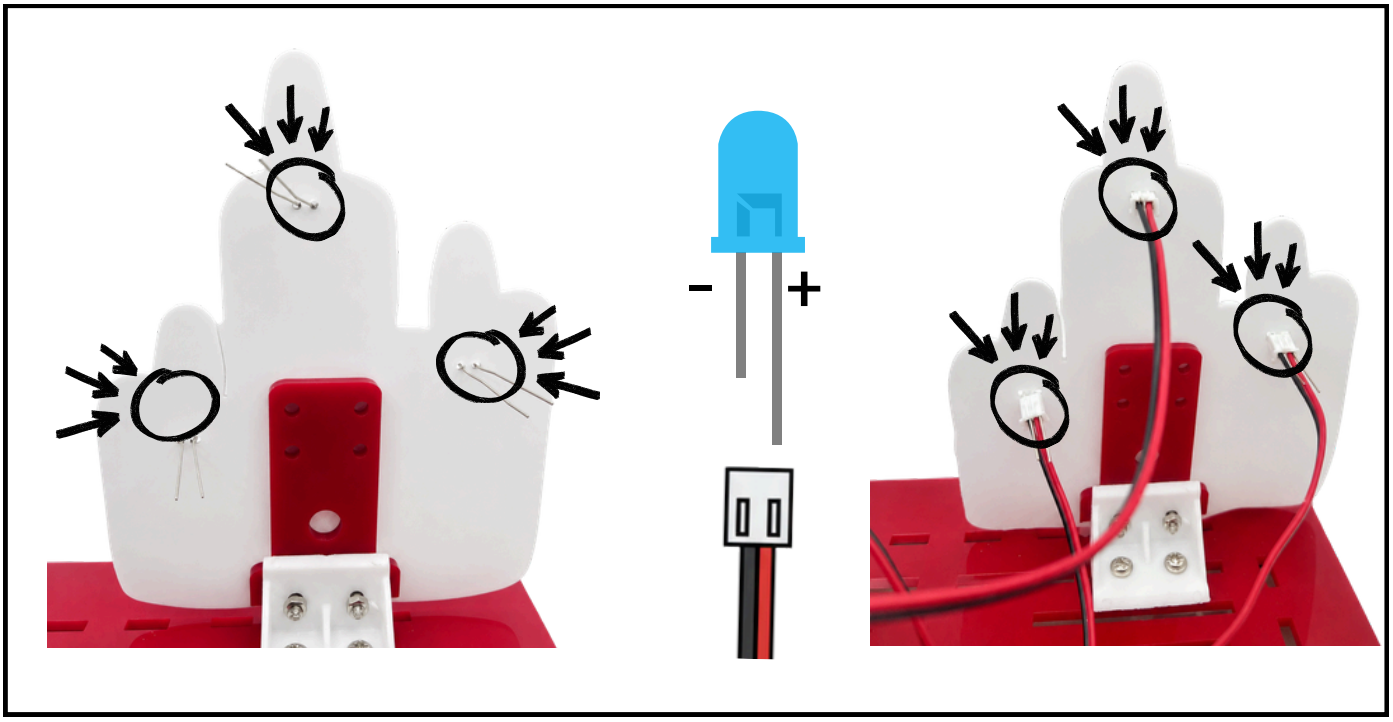
Step 2 - Insert 3mm bolt into template frontside



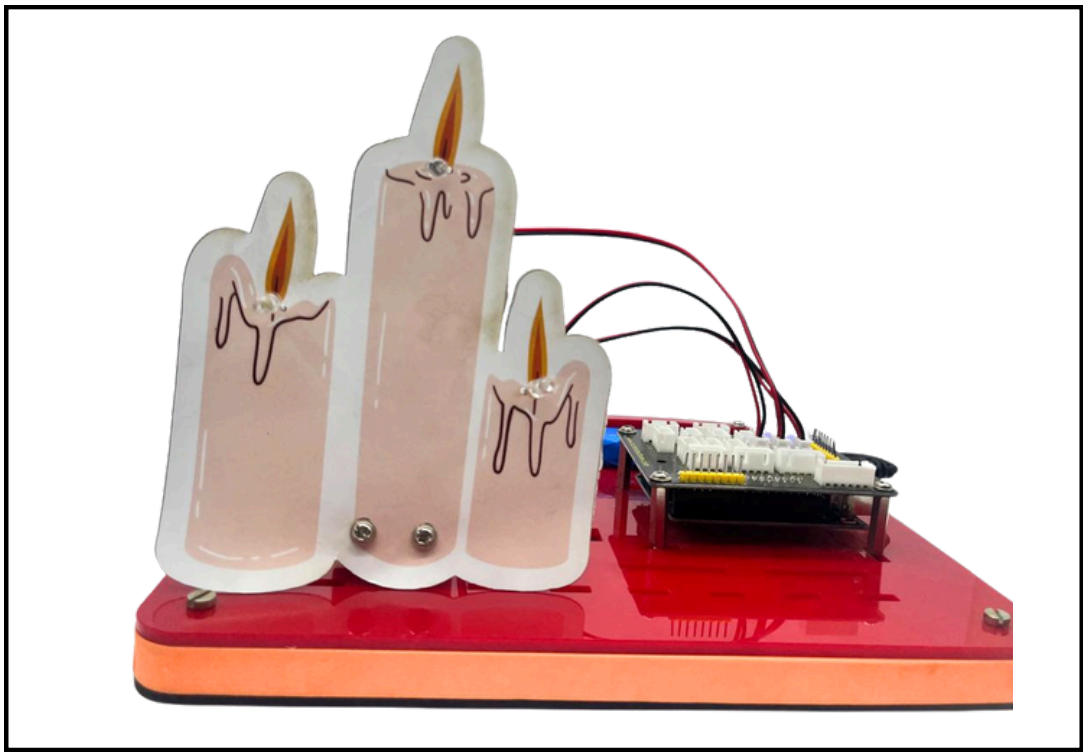
Step 3 - Attach small stand into the bolt as shown in the figure



Step 4 - Take three DIP LEDs and insert them into the template front side and connect the 2 pin rmc to leds from back side of template



upload code and observe the pattern in template



code

```
#define LED1 D5
#define LED2 D6
#define LED3 D7

void setup() {
 pinMode(LED1, OUTPUT);
 pinMode(LED2, OUTPUT);
 pinMode(LED3, OUTPUT);
}

void loop() {
 digitalWrite(LED1, HIGH);
 digitalWrite(LED2, LOW);
 digitalWrite(LED3, HIGH);
 delay(300);
```

```
digitalWrite(LED1, LOW);
 digitalWrite(LED2, HIGH);
 digitalWrite(LED3, LOW);
 delay(300);

 digitalWrite(LED1, HIGH);
 digitalWrite(LED2, HIGH);
 digitalWrite(LED3, HIGH);
 delay(200);

 digitalWrite(LED1, LOW);
 digitalWrite(LED2, LOW);
 digitalWrite(LED3, LOW);
 delay(200);
}
```